

How Our System Works

The complete process From truck arrival to optimized warehouse, told as a story

THE STARTING POINT

The Problem We're Solving

REALITY TODAY

Warehouses Operate Blind

Every day, 20 to 35 trucks arrive at a warehouse with 8 dock doors. Some show up early, some an hour late, some don't come at all. When multiple trucks arrive at once, they wait in the yard, burning money. Meanwhile inside, tasks pile up unevenly: one zone has 50 tasks and workers are overwhelmed, another has 5 and workers are idle. Nobody sees the full picture in real time.

💡 Think of an airport with no air traffic control. Planes land whenever they want, some gates are packed, others empty. Chaos.

FOUNDATION

Step 1: The Data

DATA LAYER

6 Months of Warehouse Operations

The system reads historical warehouse data: 3,500 dock appointments, 16,000 task records, carrier performance profiles, exception logs, and wave plans. We built synthetic data with real operational patterns baked in. Some carriers are chronically late, mornings have congestion peaks, certain zones get overloaded.

💡 Like a flight simulator with real turbulence patterns. We train on realistic conditions so when real data arrives, we're ready.

Data-agnostic: swap CSVs when real data arrives, zero code changes

PHASE 1

Agent 1: Dock Scheduling

STEP 1 of 6

Load Today's Dock Schedule

Agent 1 pulls every appointment for the day: which carrier, what time, which dock door, shipment size, priority level. It sees the full picture of what's supposed to happen today.

STEP 2 of 6

Profile Every Carrier's Track Record

For each carrier showing up today, the agent checks their historical reliability: on-time rate, average delay when late, no-show rate, typical unload duration. Tags each as **high**, **medium**, or **low** reliability.

💡 Like checking a restaurant's reviews before booking. "SwiftHaul: 91% on-time, great."
"DiscountFreight: 42% on-time, chronically late." Now you know who to watch.

STEP 3 of 6

Predict When Trucks Will Actually Arrive

Using carrier profiles, the agent predicts real arrival times. Not the schedule, but what will actually happen. Formula: **Predicted Arrival = Scheduled Time + (Avg Delay × Late Probability)**. A carrier scheduled for 9 AM that's late 55% of the time with a 40-min average delay? Predicted arrival: 9:22 AM.

STEP 4 of 6

Detect Congestion Windows

Divides the day into 1-hour windows. Counts how many trucks will occupy docks in each window. With 8 doors, if 9 trucks need a dock simultaneously, that's a **CRITICAL** window. Flags every window as Normal, Warning (75%+), or Critical (100%+).

💡 Like a traffic heat map on Google Maps. Green = flowing. Yellow = slowing. Red = gridlocked.
Now you can plan your route.

STEP 5 of 6

Recommend Rescheduling

Finds unreliable carriers sitting in critical windows and proposes moving them to quieter time slots. Only moves flexible shipments and never touches hot/priority loads. Each recommendation includes the reason and projected impact.

Example: "Move DiscountFreight from 9:00 to 10:30 AM. Reduces 9 AM window from 100% to 88%."

STEP 6 of 6

Show Before / After Impact

Simulates what happens if the recommendations are followed. Shows the operator a clear comparison: wait times, congestion windows, and dock utilization, before and after.

Result: Carrier wait time 14.7 min down to 7.8 min (-47%)



Cross-Agent Handoff

Agent 1's congestion predictions flow into Agent 2.
"Expect a surge at 9 AM. Zone A will be hit."

PHASE 2

Agent 2: Task Prioritization

STEP 1 of 7

Load the Task Queue

Pulls every pending task across all warehouse zones: unloading, putaway, QC checks, replenishment. Each task has a zone, type, creation time, and SLA deadline. Typical day: 150 to 200 tasks.

STEP 2 of 7

Measure Zone Balance

Counts tasks per zone and calculates the **Coefficient of Variation (CV)**, a single number measuring how uneven the workload is. CV of 0 = perfectly balanced. CV of 0.84 = severely imbalanced (one zone drowning, another empty).

💡 Like a restaurant where one waiter has 12 tables and another has 2. The CV tells you how unfair the split is.

STEP 3 of 7

Analyze Dwell Times

Dwell time = how long a task sits idle before anyone starts working on it. A task created at 8 AM that doesn't start until 9:30 AM has 90 minutes of pure waste. The agent finds every task that's been sitting too long.

STEP 4 of 7

Read Agent 1's Predictions ⚡

This is the key moment. Agent 2 reads Agent 1's congestion forecast. "Dock surge at 9 AM, Zone A will be flooded with new tasks." Agent 2 immediately starts clearing Zone A's current backlog to make room. **Cascade prevention.**

💡 Airport tower radios the gate manager: "Three planes landing in 10 minutes." The gate manager clears space before the passengers even arrive.

STEP 5 of 7

Score Every Task

Every task gets a priority score from 0 to 1 using three weighted factors:

Priority = (SLA Urgency × 40%) + (Dwell Time × 30%) + (Zone Balance × 30%)

Feeding an order due in 2 hours? High SLA score. Been sitting idle for 90 min? High dwell score. In an overloaded zone? High balance score. The agent ranks all tasks from most to least urgent.

STEP 6 of 7

Isolate Exception Tasks

Tasks where something went wrong (damaged goods, wrong SKU, missing paperwork) can't be processed normally and block the queue if left mixed in. The agent pulls them into a separate exception queue and routes them to a supervisor. Healthy tasks flow unblocked.

STEP 7 of 7

Show Optimized Queue + Impact

Presents the reordered task list with scores and reasoning. Compares the current first-in-first-out approach against the optimized ranking. Shows projected improvements across all zones.

Result: Zone balance CV 0.841 down to 0.336 (-60%) | Dwell time -30% | Throughput +12%

PHASE 3

Combined Output: The Dashboard

Three-Tab Interactive Dashboard

Tab 1, Dock Scheduling: Congestion heatmap across the day, rescheduling recommendations with reasoning, carrier reliability profiles.

Tab 2, Task Prioritization: Zone workload bars, reprioritized task list with scores, dwell time analysis, exception queue.

Tab 3, Impact Summary: All KPIs with before/after cards and percentage change. One glance tells you the value.

💡 Like a pilot's cockpit display. Everything you need to make decisions, nothing you don't.

📊 Projected Impact (Tested on 6 Months of Data)

CARRIER WAIT TIME

~~14.7 min~~ → **7.8 min** -47%

INBOUND TO PUTAWAY CYCLE

~~62.5 min~~ → **38.8 min** -38%

ZONE BALANCE (CV)

~~0.841~~ → **0.336** -60%

EXCEPTION RESOLUTION

~~216 min~~ → **86 min** -60%

✂ **The Key Innovation:** Agent 1 predicts dock congestion, tells Agent 2, and Agent 2 pre-clears the impacted zone BEFORE the surge hits. This **cascade prevention** is why we use two agents instead of one. A single system reacts after the damage. Two coordinated agents prevent it.

🔒 **Advisory Mode:** Both agents recommend, humans approve. No truck gets moved, no task gets reordered without a human saying yes.